3.1.2 Amount of substance: Sheet 10

The ideal gas equation · 16 marks · about 25 minutes

Key information

- **pV = nRT**: p in Pa, V in m^3, T in K, R = 8.31 J K^{-1} mol^{-1}
- Unit conversions: kPa × 1000 → Pa; $cm^3 \times 10^{-6} \rightarrow m^3$; $dm^3 \times 10^{-3} \rightarrow m^3$; °C + 273 → K.
- Rearrange first: n = pV ÷ RT, V = nRT ÷ p, M_r = mass ÷ n.
- At the same temperature and pressure, gas volumes are in the same ratio as moles (use the equation coefficients).

Section A: Self-quiz

01 What is the mass, in kg, of 2000 m^3 of carbon monoxide gas at 100 kPa and 25 °C?
R = 8.31 J K^{-1} mol^{-1}

[1 mark]

A 2.261
B 2.696
C 2261
D 2696

02 At temperature T and pressure P, 0.287 g of nitrogen has a volume of 25.0 cm^3
Which quantity of substance at temperature T and pressure P contains the same number of atoms as 0.287 g of nitrogen in 25.0 cm^3?

[1 mark]

A 25.0 cm^3 of argon
B 1.312 g of sulfur dioxide
C 0.287 g of carbon dioxide
D 25.0 cm^3 of oxygen

03 A 385 cm^3 sample of carbon dioxide at 100 kPa and 25 °C was mixed with 2.89×10^{-2} mol of argon. The gas constant, R = 8.31 J K^{-1} mol^{-1}
What is the mole fraction of carbon dioxide in the mixture?

[1 mark]

A 0.35
B 0.46
C 0.54
D 0.65

04 A 30 cm^3 sample of nitrogen was reacted with a 60 cm^3 sample of fluorine according to the equation
$\frac{1}{2}N_2(g) + 1\frac{1}{2}F_2(g) \rightarrow NF_3(g)$
What is the volume of the gas mixture after the reaction, at constant temperature and pressure?

[1 mark]

A 20 cm^3
B 30 cm^3
C 40 cm^3
D 50 cm^3

05 Alkanes have the general formula C_nH_{2n+2}. Alkanes undergo complete combustion in a plentiful supply of oxygen.

$C_nH_{2n+2} + x\ O_2 \rightarrow n\ CO_2 + (n+1)\ H_2O$

Determine x in terms of n

[1 mark]

Section B: Exam questions

06 M is a Group 2 metal that forms the nitrate $M(NO_3)_2$

0.320 g of $M(NO_3)_2$ is heated strongly and decomposes completely.

$2\ M(NO_3)_2(s) \rightarrow 2\ MO(s) + 4\ NO_2(g) + O_2(g)$

The mixture of gases formed has a volume of 225 cm^3 at 450 °C and 101 000 Pa

Determine the M_r of $M(NO_3)_2$

Identify M.

The gas constant, R = 8.31 $J\ K^{-1}\ mol^{-1}$

[5 marks]

07 A student determines the relative molecular mass (M_r) of a volatile liquid A by injecting a sample of A into a gas syringe in an oven. At the temperature of the oven, liquid A vaporises.

Table 1

Mass of fine needle syringe and contents before injecting	11.295 g
Mass of fine needle syringe and contents after injecting	10.835 g
Volume reading on gas syringe before injecting	0.0 cm^3
Volume reading on gas syringe after injecting	178.0 cm^3
Pressure of gas in syringe	100 kPa
Temperature of oven	120 °C

07.1 Calculate the M_r of A.

Give your answer to 3 significant figures.

The gas constant, R = 8.31 $J\ K^{-1}\ mol^{-1}$

[4 marks]

07.2 The student noticed that some of the liquid injected into the gas syringe did not vaporise.

Explain the effect that this has on the M_r calculated by the student.

[2 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	**C**	Source: 7404/1 Jun 2025 Q09	1
02	**D**	Source: 7404/1 Jun 2025 Q22	1
03	**A**	Source: 7405/3 Jun 2017 Q28	1
04	**D**	Source: 7404/1 Jun 2018 Q13	1
05	(3n + 1) ÷ 2 OR 1.5n + 0.5	Source: 7404/2 Jun 2022 Q04.3 Allow other correct expressions	1
06	M1 V = 225 × 10^{-6} m^3; T = 723 K M2 n = pV ÷ RT = (101 000 × 225 × 10^{-6}) ÷ (8.31 × 723) = 3.78 × 10^{-3} mol M3 n $M(NO_3)_2$ = M2 × ⅖ = 1.51 × 10^{-3} mol M4 M_r = 0.320 ÷ M3 = 211.9 (allow 211.5 to 212) M5 A_r(M) = 211.9 − 124.0 = 87.9, so M is Sr	Source: 7404/1 Jun 2023 Q04 M1 converts units of V and T M5 consequential to M_r value on answer line: must be a Group 2 metal	5
07.1	M1 n = pV ÷ RT M2 n = 100 000 × (178 ÷ 1 000 000) ÷ (8.31 × (273 + 120)) M3 n = 5.45 × 10^{-3} mol M4 M_r = 0.460 ÷ 5.45 × 10^{-3} = 84.4	Source: 7404/1 Nov 2020 Q08.1–08.2 M1 for rearrangement; M2 for three unit conversions M4 answer must be to 3 sf	4
07.2	M1 Calculated M_r value would be greater than actual M2 A lower volume would have been recorded / mass evaporated less than mass of liquid / lower moles calculated / mass recorded higher than mass of gas	M_r = mass ÷ moles, so dividing by too small a value of moles gives a larger M_r than expected M2 dependent on correct M1	2